

Lab 2: Jira — Agile Backlog & Sprint Simulation

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Section: F

Problem Statement: #25 — Municipal Solid Waste Route Optimizer

Jira Project: PES1UG24AM325 (key: SCRUM)

Functional & Non-Functional Requirements (from Lab 1)

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall generate optimized daily collection routes including only bins that have exceeded an 80% fill threshold.	High	Route excludes bins <80% full unless manually tagged. Fail: Overflowing bin omitted from driver manifest.	Ensures trucks are dispatched efficiently, avoiding wasted trips to non-full bins while preventing overflow.
FR-002	Functional	The system shall ingest fill-level telemetry from all smart bin sensors at least once every 15 minutes and update each bin's status accordingly.	High	Every bin's fill-level record has a timestamp no older than 15 min during operating hours. Fail: Any bin's timestamp exceeds 15 min with no logged sensor fault.	Route optimization (FR-001) depends on current data — stale readings cause missed or wasted pickups.
FR-003	Functional	The system	High	Bin status	Closes the

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
		shall allow a Truck Driver to mark a bin as collected via the mobile app, updating the bin's status to "Empty" and recording a collection timestamp.		changes to Empty and timestamp logs within 5 seconds of driver confirmation. Fail: Bin remains flagged as pending after confirmation.	operational loop so fill-level data stays accurate for the next route cycle.
FR-004	Functional	The system shall allow a Citizen to submit a missed-pickup complaint specifying the bin location/ID, an optional photo, and a description.	Medium	Submitted complaint appears in the Supervisor's queue with a unique complaint ID within 1 minute. Fail: Complaint not received, or lacks a bin location reference.	Gives citizens a channel to report service failures, supporting accountability and service-quality tracking.
FR-005	Functional	The system shall allow a Sanitation Supervisor to manually tag a bin below the 80% threshold for inclusion in the next optimized route.	Medium	Tagged bin appears in the next generated route's manifest regardless of fill %. Fail: Tagged bin is excluded from the next route.	Handles edge cases (e.g. citizen complaint, event) that automatic thresholding can't anticipate.
NFR-001	Non-Functional	Route recalculation for a fleet of	High	Benchmarking tests confirm target	Supervisors need near-real-time

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
		50 trucks across 10,000 bins shall execute in under 45 seconds.		latency under simulated peak load.	replanning during peak operations without operational delay.
NFR-002	Non-Functional	The bin sensor data ingestion pipeline and supervisor dashboard shall maintain 99.5% uptime, measured monthly.	High	Uptime monitoring logs show $\geq 99.5\%$ availability over a rolling 30-day window. Fail: Any monthly measurement falls below 99.5%.	Route planning and complaint handling are daily-critical operations; downtime directly delays citywide waste collection.

Epics

Epic	Covers
Route Generation & Optimization	FR-001, FR-005, NFR-001
Bin Sensor Data Ingestion & Monitoring	FR-002, NFR-002
Driver Collection & Mobile Operations	FR-003
Citizen Complaint Management	FR-004

User Stories

Epic	Title	Req	Priority	SP
Route Gen.	Fetch Latest Bin Fill-Level Data	FR-001	High	3
Route Gen.	Filter Bins Above 80% Threshold	FR-001	High	2
Route Gen.	Generate & Preview Optimized Routes	FR-001	High	8
Route Gen.	Publish Route	FR-001	High	3

Epic	Title	Req	Priority	SP
	Manifest to Drivers			
Route Gen.	Manually Tag Bin Below Threshold	FR-005	Medium	3
Route Gen.	Include Tagged Bins in Route Calculation	FR-005	Medium	2
Route Gen.	Meet Route Recalculation Performance Target	NFR-001	High	5
Route Gen.	Benchmark Route Engine Under Peak Load	NFR-001	High	3
Sensor Ingestion	Ingest Sensor Telemetry Every 15 Minutes	FR-002	High	5
Sensor Ingestion	Update Bin Status from Telemetry	FR-002	High	3
Sensor Ingestion	Flag Stale Sensor Data	FR-002	High	3
Sensor Ingestion	Maintain 99.5% Uptime for Ingestion & Dashboard	NFR-002	High	5
Sensor Ingestion	Monitor and Log System Uptime	NFR-002	High	3
Driver Mobile	Mark Bin as Collected	FR-003	High	3
Driver Mobile	Log Collection Timestamp	FR-003	High	2
Citizen Complaints	Submit Missed-Pickup Complaint	FR-004	Medium	3
Citizen Complaints	Generate Unique Complaint ID	FR-004	Medium	2
Citizen Complaints	View Complaint Queue	FR-004	Medium	3

Sprint Plan

Total: 61 story points, 18 stories, across 2 sprints.

Sprint 1 — 34 points — Route Generation + Driver Mobile: all 8 Route Generation & Optimization stories + both Driver Collection & Mobile Operations stories. Delivers the core loop: a Supervisor can generate/dispatch an optimized route and a Driver can mark bins collected.

Sprint 2 — 27 points — Sensor Ingestion + Citizen Complaints: all 5 Bin Sensor Data Ingestion & Monitoring stories + all 3 Citizen Complaint Management stories. Rounds out the system with the data feed powering routes and the citizen-facing complaint channel.

Jira Screenshots

Paste each screenshot directly under its heading below, in this order.

1. Backlog — Epics, Stories, Priority & Story Points

Spaces
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Summary Backlog Board Development Timeline Docs Forms +

Search backlog Filter

Backlog (18 work items) 61 0 0 Create sprint

Item	Label	Category	Status	Points	Priority	Assignee
SCRUM-41	Fetch Latest Bin Fill-Level Data	Route Generation ...	To Do	3	High	Assignee
SCRUM-42	Filter Bins Above 80% Threshold	Route Generation ...	To Do	2	High	Assignee
SCRUM-43	Generate & Preview Optimized Routes	Route Generation ...	To Do	8	High	Assignee
SCRUM-44	Publish Route Manifest to Drivers	Route Generation ...	To Do	3	High	Assignee
SCRUM-45	Manually Tag Bin Below Threshold	Route Generation ...	To Do	3	Medium	Assignee
SCRUM-46	Include Tagged Bins in Route Calculation	Route Generation ...	To Do	2	Medium	Assignee
SCRUM-47	Meet Route Recalculation Performance Target	Route Generation ...	To Do	5	High	Assignee
SCRUM-48	Benchmark Route Engine Under Peak Load	Route Generation ...	To Do	3	High	Assignee
SCRUM-49	Ingest Sensor Telemetry Every 15 Minutes	Bin Sensor Data In...	To Do	5	High	Assignee
SCRUM-50	Update Bin Status from Telemetry	Bin Sensor Data In...	To Do	3	High	Assignee

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SCRUM-49	Ingest Sensor Telemetry Every 15 Minutes	Bin Sensor Data In...	To Do	5	^	
SCRUM-50	Update Bin Status from Telemetry	Bin Sensor Data In...	To Do	3	^	
SCRUM-51	Flag Stale Sensor Data	Bin Sensor Data In...	To Do	3	^	
SCRUM-52	Maintain 99.5% Uptime for Ingestion & Dashboard	Bin Sensor Data In...	To Do	5	^	
SCRUM-53	Monitor and Log System Uptime	Bin Sensor Data In...	To Do	3	^	
SCRUM-54	Mark Bin as Collected	Driver Collection &...	To Do	3	^	
SCRUM-55	Log Collection Timestamp	Driver Collection &...	To Do	2	^	
SCRUM-56	Submit Missed-Pickup Complaint	Citizen Complaint ...	To Do	3	=	
SCRUM-57	Generate Unique Complaint ID	Citizen Complaint ...	To Do	2	=	
SCRUM-58	View Complaint Queue	Citizen Complaint ...	To Do	3	=	

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2. Sprint 1 — Active Sprint board

SCRUM Sprint 1 Add dates (10 work items) 34 0 0 Start sprint ...

SCRUM-41	Fetch Latest Bin Fill-Level Data	Route Generation ...	To Do	3	^	
SCRUM-42	Filter Bins Above 80% Threshold	Route Generation ...	To Do	2	^	
SCRUM-43	Generate & Preview Optimized Routes	Route Generation ...	To Do	8	^	
SCRUM-44	Publish Route Manifest to Drivers	Route Generation ...	To Do	3	^	
SCRUM-45	Manually Tag Bin Below Threshold	Route Generation ...	To Do	3	=	
SCRUM-46	Include Tagged Bins in Route Calculation	Route Generation ...	To Do	2	=	
SCRUM-47	Meet Route Recalculation Performance Target	Route Generation ...	To Do	5	^	
SCRUM-48	Benchmark Route Engine Under Peak Load	Route Generation ...	To Do	3	^	
SCRUM-54	Mark Bin as Collected	Driver Collection &...	To Do	3	^	
SCRUM-55	Log Collection Timestamp	Driver Collection &...	To Do	2	^	

Spaces

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To Do 10

Fetch Latest Bin Fill-Level Data

Parent

Route Generation & Optimization

SCRUM-41 3

Filter Bins Above 80% Threshold

Parent

Route Generation & Optimization

SCRUM-42 2

Generate & Preview Optimized Routes

Parent

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In Progress 0

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In Review 0

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Done 0

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To Do 1

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In Progress 12

Benchmark Route Engine Under Peak Load

Parent

Route Generation & Optimization

SCRUM-48 3

Meet Route Recalculation Performance Target

Parent

Route Generation & Optimization

SCRUM-47 5

Generate & Preview Optimized Routes

Parent

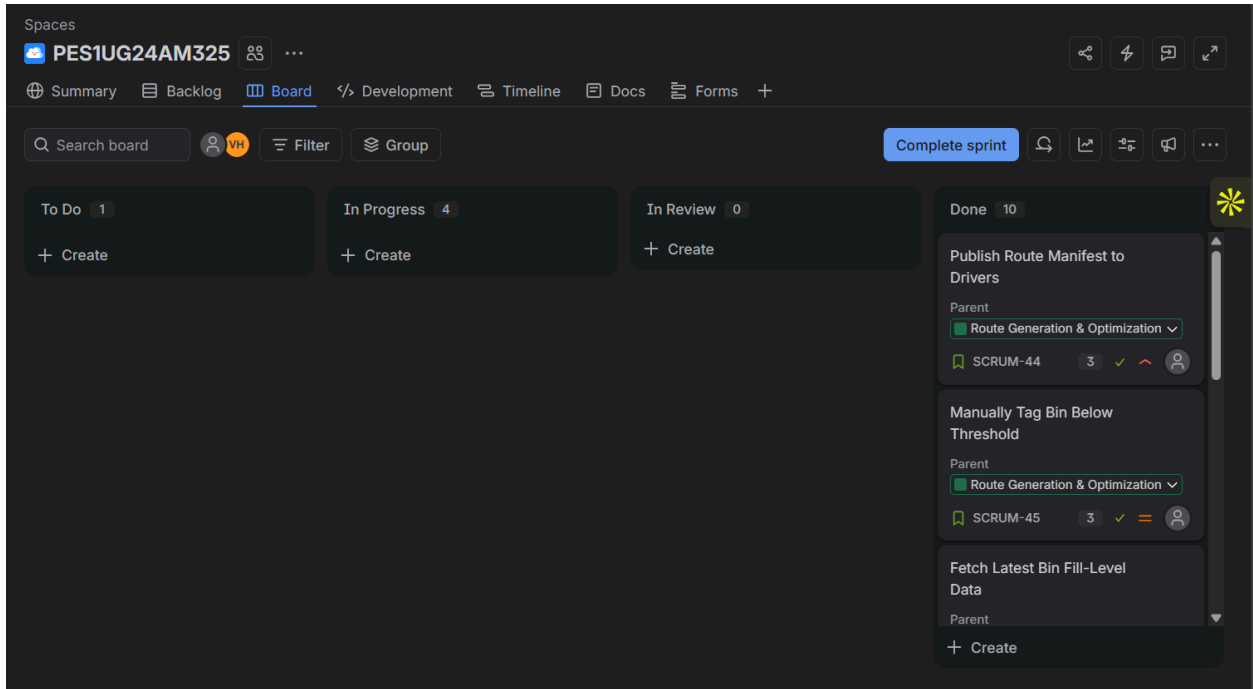
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In Review 0

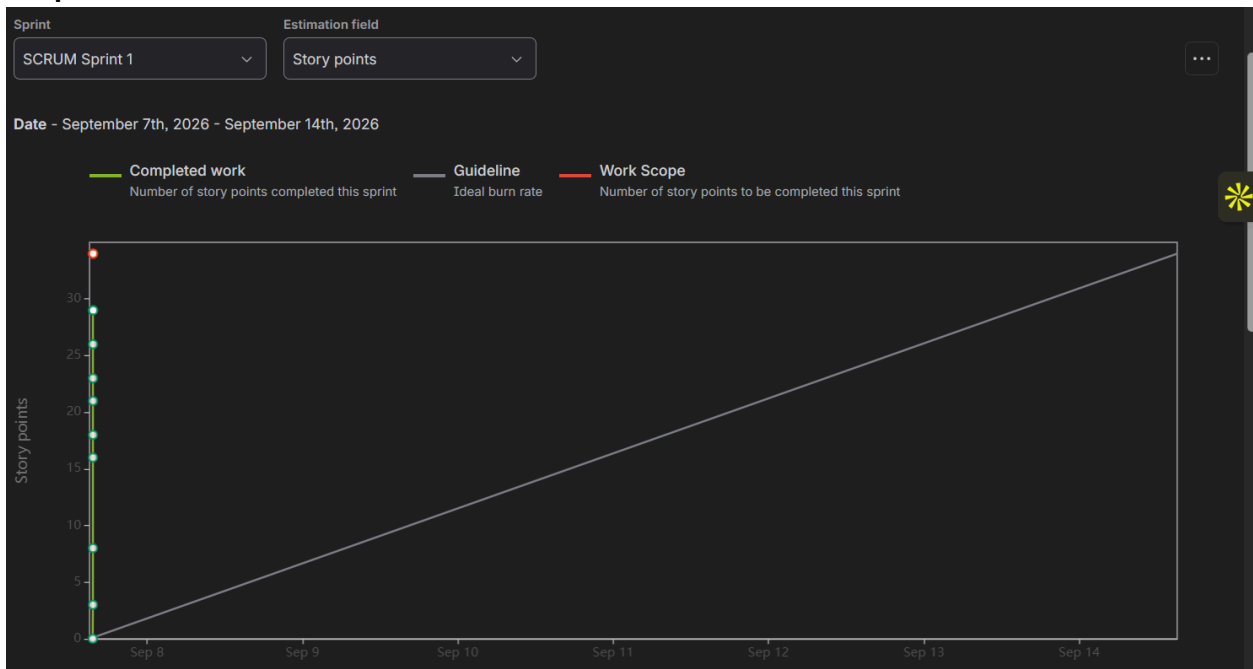
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Done 0

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3. Sprint 1 — Burndown Chart



Date	Event	Work item	Completed	Scope
Mon, Sep 07 2026, 3:07pm	Sprint started	SCRUM-42 Filter Bins Above 80% Threshold SCRUM-41 Fetch Latest Bin Fill-Level Data SCRUM-43 Generate & Preview Optimized Routes SCRUM-44 Publish Route Manifest to Drivers SCRUM-46 Include Tagged Bins in Route Calculation SCRUM-45 Manually Tag Bin Below Threshold SCRUM-48 Benchmark Route Engine Under Peak Load SCRUM-47 Meet Route Recalculation Performance Target SCRUM-54 Mark Bin as Collected SCRUM-55 Log Collection Timestamp	0	34
Mon, Sep 07 2026, 3:08pm	Work item completed	SCRUM-48 Benchmark Route Engine Under Peak Load	0 → 3	34
Mon, Sep 07 2026, 3:09pm	Work item completed	SCRUM-47 Meet Route Recalculation Performance Target	3 → 8	34
Mon, Sep 07 2026, 3:09pm	Work item completed	SCRUM-43 Generate & Preview Optimized Routes	8 → 16	34
Mon, Sep 07 2026, 3:09pm	Work item completed	SCRUM-42 Filter Bins Above 80% Threshold	16 → 18	34
Mon, Sep 07 2026, 3:09pm	Work item completed	SCRUM-41 Fetch Latest Bin Fill-Level Data	18 → 21	34
Mon, Sep 07 2026, 3:09pm	Work item completed	SCRUM-46 Include Tagged Bins in Route Calculation	21 → 23	34
Mon, Sep 07 2026, 3:09pm	Work item completed	SCRUM-45 Manually Tag Bin Below Threshold	23 → 26	34
Mon, Sep 07 2026, 3:09pm	Work item completed	SCRUM-44 Publish Route Manifest to Drivers	26 → 29	34
Mon, Sep 07 2026, 3:10pm	Sprint completed	SCRUM-42 Filter Bins Above 80% Threshold SCRUM-41 Fetch Latest Bin Fill-Level Data SCRUM-43 Generate & Preview Optimized Routes SCRUM-44 Publish Route Manifest to Drivers SCRUM-46 Include Tagged Bins in Route Calculation SCRUM-45 Manually Tag Bin Below Threshold SCRUM-48 Benchmark Route Engine Under Peak Load SCRUM-47 Meet Route Recalculation Performance Target SCRUM-54 Mark Bin as Collected SCRUM-55 Log Collection Timestamp	29	34

4. Sprint 2 — Active Sprint board

SCRUM Sprint 2 Add dates (8 work items) 27 0 0 Start sprint ...	
SCRUM-49 Ingest Sensor Telemetry Every 15 Minutes Bin Sensor Data In... To Do 5	
SCRUM-50 Update Bin Status from Telemetry Bin Sensor Data In... To Do 3	
SCRUM-51 Flag Stale Sensor Data Bin Sensor Data In... To Do 3	
SCRUM-52 Maintain 99.5% Uptime for Ingestion & Dashboard Bin Sensor Data In... To Do 5	
SCRUM-53 Monitor and Log System Uptime Bin Sensor Data In... To Do 3	
SCRUM-56 Submit Missed-Pickup Complaint Citizen Complaint ... To Do 3	
SCRUM-57 Generate Unique Complaint ID Citizen Complaint ... To Do 2	
SCRUM-58 View Complaint Queue Citizen Complaint ... To Do 3	
+ Create	

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Complete sprint

To Do 10

Mark Bin as Collected

Parent

Driver Collection & Mobile Oper...

SCRUM-54 3

Log Collection Timestamp

Parent

Driver Collection & Mobile Oper...

SCRUM-55 2

Ingest Sensor Telemetry Every 15 Minutes

Parent

Bin Sensor Data Ingestion & Mo...

In Progress 0

In Review 0

Done 0

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Complete sprint

To Do 0

In Progress 10

Submit Missed-Pickup Complaint

Parent

Citizen Complaint Management

SCRUM-56 3

Monitor and Log System Uptime

Parent

Bin Sensor Data Ingestion & Mo...

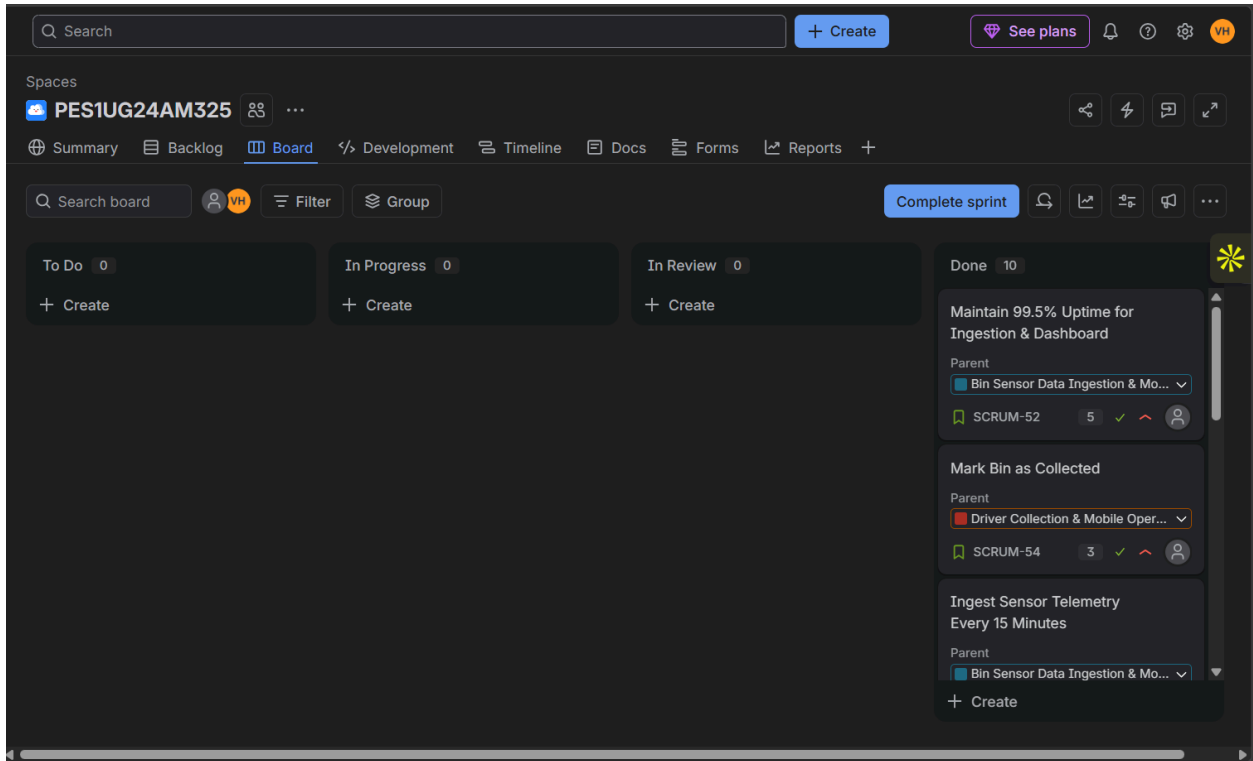
SCRUM-53 3

Generate Unique Complaint ID

Parent

In Review 0

Done 0



5. Sprint 2 — Burndown Chart



Date	Event	Work item	Completed	Scope
Mon, Sep 07 2026, 3:13pm	Sprint started	SCRUM-49 Ingest Sensor Telemetry Every 15 Minutes SCRUM-50 Update Bin Status from Telemetry SCRUM-51 Flag Stale Sensor Data SCRUM-52 Maintain 99.5% Uptime for Ingestion & Dashboard SCRUM-53 Monitor and Log System Uptime SCRUM-56 Submit Missed-Pickup Complaint SCRUM-57 Generate Unique Complaint ID SCRUM-58 View Complaint Queue SCRUM-54 Mark Bin as Collected SCRUM-55 Log Collection Timestamp	0	32
Mon, Sep 07 2026, 3:15pm	Work item completed	SCRUM-56 Submit Missed-Pickup Complaint	0 → 3	32
Mon, Sep 07 2026, 3:15pm	Work item completed	SCRUM-53 Monitor and Log System Uptime	3 → 6	32
Mon, Sep 07 2026, 3:15pm	Work item completed	SCRUM-57 Generate Unique Complaint ID	6 → 8	32
Mon, Sep 07 2026, 3:15pm	Work item completed	SCRUM-58 View Complaint Queue	8 → 11	32
Mon, Sep 07 2026, 3:15pm	Work item completed	SCRUM-50 Update Bin Status from Telemetry	11 → 14	32
Mon, Sep 07 2026, 3:15pm	Work item completed	SCRUM-55 Log Collection Timestamp	14 → 16	32
Mon, Sep 07 2026, 3:15pm	Work item completed	SCRUM-49 Ingest Sensor Telemetry Every 15 Minutes	16 → 21	32
Mon, Sep 07 2026, 3:15pm	Work item completed	SCRUM-54 Mark Bin as Collected	21 → 24	32
Mon, Sep 07 2026, 3:15pm	Work item completed	SCRUM-52 Maintain 99.5% Uptime for Ingestion & Dashboard	24 → 29	32
Mon, Sep 07 2026, 3:15pm	Work item completed	SCRUM-52 Maintain 99.5% Uptime for Ingestion & Dashboard	24 → 29	32
Mon, Sep 07 2026, 3:15pm	Work item completed	SCRUM-51 Flag Stale Sensor Data	29 → 32	32
Mon, Sep 07 2026, 3:15pm	Sprint completed	SCRUM-49 Ingest Sensor Telemetry Every 15 Minutes SCRUM-50 Update Bin Status from Telemetry SCRUM-51 Flag Stale Sensor Data SCRUM-52 Maintain 99.5% Uptime for Ingestion & Dashboard SCRUM-53 Monitor and Log System Uptime SCRUM-56 Submit Missed-Pickup Complaint SCRUM-57 Generate Unique Complaint ID SCRUM-58 View Complaint Queue SCRUM-54 Mark Bin as Collected SCRUM-55 Log Collection Timestamp	32	32

Reflection Questions

Q1. Did your estimations reflect the actual effort?

Yes. The story points reasonably represented the complexity of the tasks, and all planned work was completed successfully.

Q2. Was your backlog well-prioritized?

Yes. The user stories were logically grouped under relevant epics and distributed across three sprints in a structured order.

Q3. How did your simulated sprint align with your plan?

The sprints followed the planned workflow, with tasks moving from To Do → In Progress → Done, and all planned items were completed.

Q4. What insights did the burndown chart give about your team's capacity?

The burndown charts showed the remaining story points reducing to zero, indicating that the planned sprint workload was successfully completed.